

MW 09/11/17

Week 6

Monday 30th October 25 min meeting with Matt. Q's on separate sheet working on paper. Also sorting logbook submission for Blackboard. 2 hours researching & typing up discrete case. I also wrote my seminar review on LaTeX to practice other types of input.

Wednesday 1st November 30 mins researching and making notes on discrete case.

Friday 3rd November 30 mins reading up continuous case. couldn't make notes on the train but it makes sense.

Saturday 4th November 2.5 hrs mainly typing up notes made previously. Don't think I'll get much more done this week due to another difficult assignment.

Going forwards Not a good effort this week. I'll spend next Wednesday working on the project & try to meet Matt later on in the week so we have more to discuss.

• When defining a source (page 43-44) Shannon does an approximation to English that I really like ~~to~~ and I would like to include in my paper. I think it's a bit long for the actual paper, can I put it in an appendix? Yes. It is too long to quote in the paper. Write a short explanation of it & refer to appendix.

• Shannon has only defined Entropy (page 50). Can I state this and use directly or should I prove it as well?

At least an outline proof for most common situations.

• For the logbook submission, do I only need to put the pages with dates or should I put everything such as my annotated Research Plan from one of the meetings, my LaTeX notes etc? Include more rather than less.

So include annotated Research Plan, LaTeX cheat sheet etc.

• If I'm using defns from Shannon's paper, should I cite at the beginning of the defn or the end? (Note for Sect. 2.1.1 & thm 2.1.3)
End is probably better but double check other papers & handbook to see what is there.

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Week 7

Monday 6th November Changed meeting to Thursday 1:30pm.
30 mins learning how to insert images using
Wrapping & picins packages.

Wednesday 8th November picins is causing issues so not using it.
4hrs typing up discrete case/researching.
Also wrote first introduction. Will need amending as I go
through. Have made notes in red of bits to add later
on. Also still need to sort examples.

Friday 10th November 1hr on brain reading the Logician & the
Engineer for some background info but
very little info I can actually use. Hopefully
as I get further through there will be more
useful information.

Thursday 9th November 25 minute supervisor meeting. Discussed
appendices & work already done.

Total 6hrs

Going forwards Being at home over the weekend
this week didn't help but next week/weekend
I should have chance to get lots of work done.

Chapter 1

Introduction

”The introduction should describe the objectives of the project, the background and review the literature.” - From the hand book. Is going to change.

We currently live in the Information Age, the age of Digitisation. In 1837, Charles Babbage envisioned the first computer when he designed his Analytical Engine, but it took over 100 years for the first electronic computer, the Atanasoff-Berry to be built in 1942. Not long after this, Claude Shannon wrote a ground breaking paper that laid down the foundations of Information Theory and therefore the mathematical basis for digitisation. This would allow the change from analogue and mechanical methods to digital that would be the defining element of the Information Age. Shannon’s paper didn’t contain designs for anything digital or the instruction regarding how to find it, he simply considered the transmission of information, and generalised the information that was transmitted over the channel. Within this paper, we will explore what technology was around before Shannon published *The Mathematical Theory of Communication*, the main points from this paper and how they have gone on to bring about the world as we know it today with smart phones, the Internet and everything else related to them.

Chapter 3

Discrete Systems

A discrete communication system is one, such as sending a telegraph, where “the message and signal are a sequence of discrete symbols” [1] Morse code is also an example of a discrete system as each character has its assigned sequence of dots and dashes and each letter is independent. Within this section we will explore the capacity of discrete channels, how to make them more efficient, how we can quantify the entropy of the system and how noise affects all of the above. Firstly, we should define a basic communication system containing a transmitter, a channel for the signal to be transmitted along and the receiver.



Figure 1. *A basic communication system.*

Once you add in noise at the channel part, the system gets slightly more complicated but that is a later section in this report.

3.1 Noiseless Systems

To begin with, we will work on discrete noiseless systems. In reality, systems are very rarely noiseless, but this will allow us to derive the basic formulae before moving on to the more useful theorems within noisy systems.

Definition 3.1.1. A **channel** is defined as either a physical connection, such as an ethernet cable, or a broadcast, such as radio waves, that is used to transmit a sequence of elements chosen from a finite source, $S_1, S_2, \dots, S_n$, from one location to another. Each symbol $S_i, 0 \leq i \leq n$ has a certain duration, t_i seconds such that t_i is not necessarily equal to $t_j, 0 \leq j \leq n$.

Definition 3.1.2. A **discrete source** is any stochastic process that chooses a sequence of symbols based on a set of probabilities. This can include languages such as English or French, but also mathematical examples where we abstractly define the stochastic process, using a discrete Markoff process, and generate our sequence of symbols from that.

It can be shown that taking a mathematical stochastic process with certain approximations, you can actually get something very similar to English with a second order word approximation. **Need to add in the appendix with this in.**

For example, the dots, dashes and spaces in Morse code are different lengths of time in the transmission.

3.1.1 Entropy

"In all energy exchanges, if no energy enters or leaves an isolated system the entropy of that system increases" Muse - Unsustainable.

Entropy is the name given to the function that we use to measure how much uncertainty there is within an information source. We define it as follows:

Definition 3.1.3. Shannon Entropy is written mathematically as

$$H = -\sum p_i \log p_i$$

where H has the following properties:

1. H is continuous in p_i
2. If $p_i = p_j \forall 0 < i, j \leq n$ H should be a monotonic increasing function of n . If events are equally likely, we have more choice and if there are more events, we have more choice.
3. If a choice is broken down into two successive choices, the original H should be the weighted sum of the individual values of H . [1]

ther way of defining Entropy is to say it is how different from the expected signal the actual signal is. So

It follows that entropy is always positive unless we are certain of the outcome, that is, $p_i = 0$ for all but one option, and this option has probability 1. Shannon Entropy is a generalisation of Hartley's information content which shows us the entropy of an information source if all of the symbols are equally likely. **Show that we can get Hartley's information from this defn for Entropy and maybe show it is an upper bound for the entropy. Can find in studentr book**

3.1.2 Encoding and Decoding

So far, we haven't dealt with the transmitter and receiver, defined above in the general communication system, from a mathematical point of view. Both parts are referred to as a *discrete transducer* and they either input a sequence of input symbols $x_1, x_2, \dots, x_n, \dots$ or output a sequence of output symbols $y_1, y_2, \dots, y_n, \dots$. A transducer may also have internal memory so the output can depend on this as well as the input. A transducer has a finite set of m possible states and the output is a function of the present state and the present input. This can be written as two functions:

$$y_n = f(x_n, \alpha_n), \quad \alpha_{n+1} = g(x_n, \alpha_n)$$

where x_n is the n th input symbol, α_n is the state of the transducer at input x_n and y_n is the output when x_n is introduced at the α_n state.

3.1.3 Fundamental Theorem for a Noiseless Channel

Theorem 3.1.4. [1] *Let a source have entropy H (bits per symbol) and a channel have a capacity C (bits per second). Then it is possible to encode the output of the source in such a way to transmit at the average rate $\frac{C}{H} - \epsilon$ symbols per second over the channel where ϵ is arbitrarily small. It is not possible to transmit at an average rate greater than $\frac{C}{H}$.*

Proof. The second part of the theorem, that $\frac{C}{H}$ cannot be exceeded can be shown if we consider that the entropy of the input per second is equal to the entropy per second of the source. This is because the transmitter must be non-singular, as the receiver is its inverse, so $H' \leq C$ and the number of symbols per second $= \frac{H'}{H} \leq \frac{C}{H}$. For the first part of the theorem, $\square$

Need to prove first part and go into more detail about this. Page 59 mtc

3.1.4 Examples

Need to put in examples of all of these. Try more up to date things though, not like Shannon used.

3.2 Noisy Systems

In most cases, transmission systems are affected by noise. This is where a signal is modified in an unwanted way by a variety of different factors. We see examples of it when a TV signal is poor and parts of the image don't show and when audio contains white noise such as when you rip a CD that has been scratched so cannot be copied completely correctly. Noise means a signal is less efficient as it is carrying information that is not required and it makes it more difficult to decode the message at the other end of the system. Digital transmission is a way of reducing noise as we know that the transmitted is either 0 or 1, transmitted at certain frequencies and if we receive frequencies that aren't the two we are expecting, we can make a good guess at which one is supposed to be received. However, there is still some guessing and we cannot be certain that we have decoded the message correctly. As a result, we would like to keep noise to a minimum as eradicating it completely is highly unlikely.

3.2.1 How do we represent a Noisy Channel?

Firstly, we need to consider how to write down this noisy channel that we are dealing with. We need to be aware that we don't know where the noise has come from, it could be either transducer or from the transmission in the middle, or a combination of the two. So the received signal may not be the same as the one that was sent from the transmitter. We can break this down into two cases, the first is that a particular transmitted signal is always affected by noise in the same way, so we can define a definite function for the received signal from the transmitted signal. We call this *distortion*. If the function has an inverse, we can, in principle, reverse the distortion in the same way we would reverse a code using a cipher.

The other case is when the change in the signal is not always the same. If we call the received signal E and we assume it is a function of the transmitted signal S and noise N , we can write it as $E = f(S, N)$. As we know nothing about the noise added to the signal, it is a chance variable that we can represent using a stochastic process. We can take a generalised version of the the noise free channel from the previous section. Assume a finite number of probabilities

$$p_{\alpha,i}(\beta, j)$$

This means that if the channel is in state α and we transmit symbol i , symbol j is received and β is the state of the channel at the end. If consecutive symbols are independently affected by the noise, there is only one state and the channel will be described by

3.2.2 Channel Capacity

$$capacity C = \text{Max}(H(x) - H_y(x))$$

3.2.3 Fundamental Theorem for a Discrete Channel with Noise

3.2.4 Example

3.2.5 Special Cases

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Week 8

Monday 13th November 30 mins reading Log & Eng but still very little useful information. Hopefully once I've written about Shannon I will be able to include some info on Bode for the historical content of my project.

Friday 17th November 25 min Supervisor meeting. Not achieved much this week due to being ill & other deadlines. Feeling positive for the weekend though. 30 mins watching a TED Talk <https://ed.ted.com/on/QNR9BK6P>. Notes are on next page.

1hr reading around breakthroughs of the info age to add further on in paper. Will detail internet, preceded by networks and possibly go into some detail about computers, from Babbage's Analytical Engine, to Atanasoff Berry, to laptop to smartphone.

Sunday 19th November 4hrs reading around the Info Age. Want to write about the maths before & after as well as Shannon's mtc paper. I have a copy of "Communication Theory of Secrecy Systems" so I can use this to do a chapter on cryptography. I have found a couple of books on the creation of the internet but I have no maths to go with that yet so it will be a small section. I have a framework for the two chapters based on mtc but I need to break down before & after & secrecy systems more. Still need to get "A Mind at Play".

Looking forwards I need to find more books/info on developments from Shannon's work. Also need to ensure there is more info in my log book.

Ted talk 30 mins

Digital communications and storage - Shannon laid groundwork for
CD/Internet/Phones/Compression bits per second.

Inv of computer

This is basis of computer circuits. Logic Machine.

Lots of work for War. Cryptography.

"Communication Theory of Secrecy Systems."

"bit" short for binary digit

Binary can be sent further as less susceptible to noise than analogue.

Changed amplifier to regenerative repeater, had same bit they started with over a mile.

Fundamental limit. (Shannon Limit/capacity)

Gave theory but didn't say how to do it.

Now within ϵ of Shannon Limit.

Compression: JPEG, MP3, ZIP files

Redundant bits in a message.

TOOK years for technology to catch up with theory.

AI. Teaching computers. Thevsew mouse learn from experience.

Built more than 30 unicyles

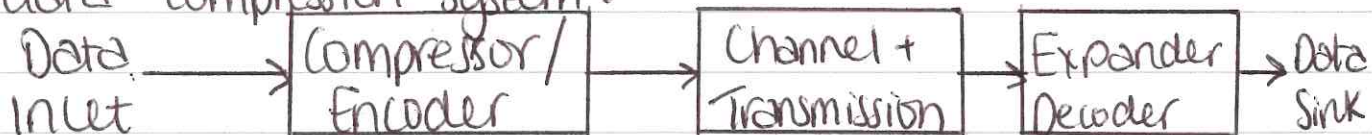
UCSD - centre of info theory

MU20/n/r

Week 9

Monday 20th November 30 minutes adding to skeleton for Security Systems & how has it impacted us chapters. I have found a couple of journals to read about it too. 25 minute supervisor meeting. Discussed how to improve logbook with more maths & examples of work done. Went through skeleton of paper. Happy with doing a small bit on what came before Info Age then going on to discuss how the internet was developed, data compression with up to date examples.

Wednesday 22nd November 1.5hrs research on data compression. I've been reading & making notes from the MPEG handbook. Focussing on audio compression as video gets more complex due to higher dimensions. Covered basic data compression system:



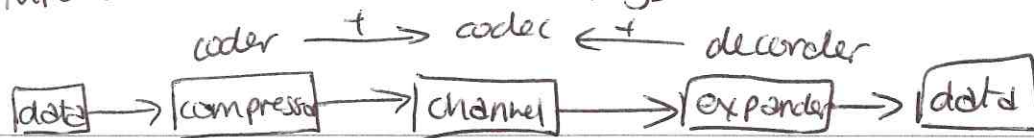
Not covered the maths yet, just doing basics. The maths may not be covered in this book. Notes are on following page from the book. LaTeX diagram is more even than this.

Thursday 23rd November 1hr I typed up a little more of the data compression info and then began writing the intro to Ch6, "How has it changed the world?" Involved reading *The Bandwagon*, 1956 by Shannon who felt Information Theory had become a fashionable topic to study & link into your studies as it had been applied to biology, psychology, linguistic, economics and other topics.

Tuesday 21st November 30mins I went through the feedback for the logbook and made a template for future logbook entries then forgot to write in the logbook.

Sunday 26th November I'm reading around how digitalisation has changed being a pilot and the mechanics of flying. Made radar better so more awareness of weather, comms are better with satellite radios now. Navigation is part of autopilot rather than done manually. What about the fact we have to put our phones on flight mode? Interfere with an aircraft's electronics? I read some interviews / have prior knowledge from VAS. Discuss with Matt on Wednesday if this would be a good way to develop paper in later chapters? Drones?

Going forwards I feel like the work I've done this week is good, just need to do more although looking at the assignments & class test I have due soon, I'm not sure this will happen.



Nearly the same info carried by a much smaller amount of data. Audio, dynamic range is reduced.

- Either reduction in bandwidth or increase in info sent, or better quality.

Lossless coding: data from decoder & original data are identical. Useful for computer programs.

Lossy coding: Original & data from decoder not same bit for bit. Not useful for programs but ok for audio-visual. Allow greater compression factors and can be done in such a way that humans find them hard to detect. This is a perceptive code. They require a model of human senses.

Entropy is how different from the expected signal the actual signal is. Can make decoders that know signals that have gone before & can use that to generate some of the current signal. Transmitter can omit parts the decoder can recreate. In theory only need to send entropy as it is the only bit that has unexpected & unpredictable info. In reality, not possible.

MORSE code is efficient in English, e is most common letter.

Try comparing morse code to scrabble letters?

Video is 4 dimensions.

Compressed data is more susceptible to errors as redundancy is removed and this is what makes codes resistant to errors.

2 types of mic, one measures changes in air pressure & one measures ~~changes~~ the air velocity as $v=0$ means silence. Pulse Code Modulation (PCM). Measure analogue wave periodically & record voltage as whole number. Then store/transmit whole number not original waveform. Whole numbers are less susceptible to drift so less likely to get errors.

Compression factor: Ratio between source data rate & channel data rate.

Chapter 6

How has this gone on to change the world as we know it?

He created a whole field in one paper. Definitely mention this. Shannon award, the Brighton conference he went to??

Once Claude Shannon had written his paper, Information Theory as a field of study was born. Shannon laid the groundwork for digital communications, something we take for granted today. His paper had such a profound effect and gained such a following in a short space of time that in 1956, Shannon wrote a paper called *The Bandwagon*. In this paper, he seemed to lament how popular Information Theory had become and how people had used the information he published. Shannon felt that people applied the maths without proper consideration of how they had actually applied it, especially in subject areas outside of the mathematics of communication, where the maths had been aimed.

However, Shannon had provided the basis for a lot of breakthroughs that we will look at in more detail in this section.

6.1 Data Compression

Does this link to Numerical Methods with the interpolation Q on Assignment 2 Sem 1?? studentr page 151.

Data compression is a huge part of our every day lives, from CD's to digital radio to the signals our TV's receive. In this section we will consider audio compression as it is simpler but the ideas can be expanded into the 4-dimension video compression as well. The basic idea of compression is that we can carry the same amount of data but using a smaller bandwidth or that we can carry data a lot more quickly using the same size of bandwidth. It is clear why this is a benefit as it will save money on the bandwidth required or on the amount of data transmitted. The compression factor is a way of measuring how much bandwidth we are saving.

Definition 6.1.1. The **compression factor** is the ratio between the source data rate and the channel data rate. [9]



Figure 2. *A basic data compression system. [9]*

There are two types of coding, both with their own merits.

Definition 6.1.2. *Lossless Coding* *is where the data that the decoder outputs and the initial input data from the data inlet are identical bit for bit.*

Lossless coding is very good for computer programs as if one bit is incorrect, such as missing one semi colon in a C++ program, the whole code won't work. It does mean that the compression factor cannot generally go above 2:1 so it is not always the cheapest option and if we can afford to lose some of the data, lossy coding will be used.

Definition 6.1.3. *Lossy Coding* *is compression where the data from the decoder and the initial data are not the same bit for bit. [9]*

Lossy coding is predominantly used in compressing audio and video data. In audio, some frequencies can be lost if there are enough other frequencies in the audio data that make it difficult for the human ear to detect the loss. This sort of coding is called perceptive and requires a model of the human senses that is fairly accurate.

In order to save more time and therefore money, we can make decoders that remember the signals that have gone before and can use that to generate some of the current signal. This means that the transmitter doesn't need to transmit part of the signal. In theory, you only need to send the entropy of a signal as it is the only unexpected and unpredictable part of the signal but in practice, this isn't possible due to noise, loss of signal over distance and other factors that mean the quality of signal is lost and that we would be missing bits of the signal after decoding. It is worth noting, however, that compressed data is more susceptible to corruption and errors as the redundant code is removed (that which we can calculate from what has gone before) and this is what makes a code resistant to errors.

6.2 Digitisation

6.3 ARPANET

6.4 Internet

YW 29/11/17

Week 10

Tuesday 28th November 1hr typing up the rest of the Data Compression notes, completed section below, notes above.
When reading B taking notes, I have a tendency to go off on a tangent B need to be aware of wasting time this way
30 mins on Progress Report. mark scheme B work required.

Wednesday 29th November 25 min Supervisor meeting.
Discussed progress and possibility of linking to flying. Advised to hang on B see how the project develops first. Need to break down Ch5 on Secrecy Systems B continue populating the rest of the report. Need to think about Midway Report
Spent an hour scanning in log book so far and ensuring it is up to date to Week 9 to save work down the line. All weeks saved separately on memory stick B One Drive.

Thursday 30th November 30 mins making preliminary notes B starting the writing up of Midway Report. Difficult to work on the train though.

Friday 1st December Spent an hour working on midway report. Need to fill in main section but unsure what to discuss within it. Need to amend the timeline for Research Plan but happy with the introduction I've written so far.

Looking forwards I've got a few assignments due next week but the final one is handed in on Thursday so I can get a few hours done over the weekend. Need to have my midway report done by Sunday so Matt can have a read for me.

Midway Report Guidance from Evgeny

Due Monday 18th December 2017, 4pm

• 1500-2000 words

• Description of what has been done so far.

• What sources have been studied

• How is Research Plan being implemented? Modifications

• Evidence of current work & finished work.

• Worked examples of theory.

• Intro contains verbal description/analysis of work, progress in implementing Research Plan & amendments.

• Main sections on particular topics of project & work from the project.

• PRESENTATION: Title, Author, Indicate supervisor, Intro, Bibliography. No abstract.

Marking criteria (10% of total mark)

Background material properly used (10)

Depth & progress on maths contents (25)

Understanding & interpretation (25)

Quality of presentation; Intro (8)

Quality of presentation; Main body/sections (8)

Quality of references (8)

Language & grammar (8)

Quality of formulae/figures etc (8)

rw 04/12/17

Week 11

Monday 4th December 25 minute supervisor meeting. Discussed handins before Christmas. I need to have a draft of the midway report done by the next meeting. I'll work on it over the weekend due to other deadlines in the week.

Friday 8th December 30 mins updating B scanning log book in. 1 hour sorting midway report & deciding which bits of report to include. Emailed Matt to clarify what goes in the bibliography for it, what I should include as I had a lot of text but nothing I feel is substantial enough to include. Done some research on morse code for examples.

Saturday 9th December 1hr typing up amendments to my action plan on the Research Plan. The bits in green are what I've changed. As they're in a table, LaTeX doesn't include this in the word count. It's approx 300 words so bear in mind when doing final word count.

Sunday 10th December ^{4 hours total} Despite the distraction of snow, I did some work on my project today. I decided not to have a separate chapter on what came before Information Theory as I can add a couple of paragraphs on this to my introduction. This means I've messed up all of my chapter numbering thus far. Before continuing with the Midway Report, I'll spend some time showing how we can take Shannon's general entropy and reduce it to the specific case of Hartley's information content for equiprobable symbols for a one character message.

Shannon Entropy is $H = -K \sum p_i \log p_i$ (Defn 2.1.3)
where p_i is the probability of the i -th symbol occurring & K is a positive constant.

If all symbols are equiprobable then $p_1 = p_2 = \dots = p_n = \frac{1}{n}$
where n is the number of symbols. If we substitute this into the definition of Shannon Entropy, we get

$$H(p_1, p_2, \dots, p_n) = -K \sum_{i=1}^n \frac{1}{n} \ln \frac{1}{n}$$

These can be equated as $n=1$ from $\dagger$ in this eg.

$$= -\frac{K}{n} \sum_{i=1}^n \ln \frac{1}{n}$$

$$= -K \ln \frac{1}{n}$$

$$= K \ln n$$

$$\ln\left(\frac{x}{y}\right) = \ln(x) - \ln(y)$$

$$\ln(1) = 0.$$

$\dagger K \ln \sum_{i=1}^n$ is Hartley's ^{measure of info} ~~function~~, first published in
Ralph V.L. Hartley, Transmission of Information, Bell Systems
Technical Journal, 7(3):535-563, July 1928.

Spent 90 minutes reading/understanding/typing this
up & ensuring I referenced my first journal correctly.

30mins Added into Midway Report a small section on
data compression (ch 5 of report). Need to try &
find some of the maths behind that at some point.

I've done a draft report now, need to ask Matt about it.

- Currently 1052 words, doesn't count the table I've used for my action plan. As this is substantial (circa 300 words) should I include it in my count? I think Turn It In does.
- Everything I've copied from my actual report is in blue. Is this acceptable, firstly? Considering the fact 25 marks go towards depth of maths used, do I need to include more maths?
- With the content used from the main report, I only have 3 references. I don't think this is enough, how many do you suggest aiming for?
- Evgeny said we should include the sources

studied. Is this implied from what I have written or do you recommend doing a section (150-250 words) on literature used?

- For this report, as it is about my progress, can I write it in first person as I did for the Research Plan or should I use the third person as I will for the final report?
- Going back to is there enough maths question. If there isn't can I go into detail about efficient coding & use Morse code as an example (this will also have more references) or should I go for a solid mathematical proof?

I haven't written my conclusion yet but I will endeavour to include a couple of more specific developments I intend to do, developing entropy for a discrete noisy channel & a continuous channel. Also the proof of the unbreakable code from Shannon.

Going forwards Next week I need to submit my second logbook which shouldn't take long as most of it is scanned and ready. I also need to complete the Midway Report to hand in on 18/12/17. Also double check with Matt if we're meeting that week or not.

Midway Report Guidance from Evgeny

Due Monday 18th December 2017, 4pm

- 1500 - 2000 words - Doesn't include table. Aim for < 1700 due to Research Plan table.
- Description of what has been done so far.
- What sources have been studied

9/12/17 ✓ How is Research Plan being implemented? Modifications

10/12/17 ✓ Evidence of current work & finished work. 2.1, 2.1.1, 5.1

✓ Worked examples of theory. 10/12 Entropy section

✓ Intro contains verbal description/analysis of work, progress in implementing Research Plan & amendments.

10/12/17 ✓ Main sections on particular topics of project & work from the project.

1/12/17 ✓ PRESENTATION: Title, Author, Indicate supervisor,

1/12/17 ✓ Intro, Bibliography. No abstract.

Marking criteria (10% of total mark)

Background material properly used (10)

Depth & progress on maths contents (25)

Understanding & interpretation (25)

Quality of presentation; Intro (8)

Quality of presentation; Main body/sections (8)

Quality of references (8)

Language & grammar (8)

Quality of formulae/figures etc (8) | This is what we've been told.

To-Do List

started 3/12/17 9/12/17 ✓ Doing separate section on Research Plan Amendments

✓ Discrete system section. ✓ Intro 10/12/17 ✓ Entropy 10/12/17

✓ Data Compression section 10/12/17

Monday 11th December 25 minute supervisor meeting. Ask the questions on the previous page RE. Midway Report. Check if meeting next week.

Next scheduled meeting is wTB January 22nd but can email if any issues before.

To-Do List from meeting

- ☐ Summarise Research Plan changes, don't need table.
- ☐ Don't use colours on actual Report.
- ☐ Include summary of a proof to show more maths
- ☐ Comment on Literature as not many references yet.
- ☐ Conclusion - include any preliminary research that is to be developed ie. using Fourier transforms.

30mins Printed out report thus far to annotate where I need to make changes. (Attached.)

Tuesday 12th December 1hr Work on midway report improvements. Details are on the print out below. I integrated the Research Plan amendments into the introduction, did all of the small grammar amendments and changed the section/subsections of the main part.

Wednesday 13th December 30mins Working on logbook. Got everything scanned in & put together into one document to submit.